

2.2 ctype.h

The ctype header is used for testing and converting characters. A control character refers to a character that is not part of the normal printing set. In the ASCII character set, the control characters are the characters from 0 (NUL) through 0x1F (US), and the character 0x7F (DEL). Printable characters are those from 0x20 (space) to 0x7E (tilde).

Functions:

```
isalnum();  
isalpha();  
iscntrl();  
isdigit();  
isgraph();  
islower();  
isprint();  
ispunct();  
isspace();  
isupper();  
isxdigit();  
tolower();  
toupper();
```

2.2.1 is... Functions

Declarations:

```
int isalnum(int character);  
int isalpha(int character);  
int iscntrl(int character);  
int isdigit(int character);  
int isgraph(int character);  
int islower(int character);  
int isprint(int character);  
int ispunct(int character);  
int isspace(int character);  
int isupper(int character);  
int isxdigit(int character);
```

The is... functions test the given character and return a nonzero (true) result if it satisfies the following conditions. If not, then 0 (false) is returned.

Conditions:

isalnum a letter (A to Z or a to z) or a digit (0 to 9)
isalpha a letter (A to Z or a to z)
iscntrl any control character (0x00 to 0x1F or 0x7F)
isdigit a digit (0 to 9)
isgraph any printing character except for the space character (0x21 to 0x7E)
islower a lowercase letter (a to z)

isprint any printing character (0x20 to 0x7E)
ispunct any punctuation character (any printing character except for space character or isalnum)
isspace a whitespace character (space, tab, carriage return, new line, vertical tab, or formfeed)
isupper an uppercase letter (A to Z)
isxdigit a hexadecimal digit (0 to 9, A to F, or a to f)

2.2.2 to... Functions

Declarations:

```
int tolower(int character);  
int toupper(int character);
```

The to... functions provide a means to convert a single character. If the character matches the appropriate condition, then it is converted. Otherwise the character is returned unchanged.

Conditions:

tolower If the character is an uppercase character (A to Z), then it is converted to lowercase (a to z)

toupper If the character is a lowercase character (a to z), then it is converted to uppercase (A to Z)

Example:

```
#include<ctype.h>  
#include<stdio.h>  
#include<string.h>  
  
int main(void)  
{  
    int loop;  
    char string[]="THIS IS A TEST";  
  
    for(loop=0;loop<strlen(string);loop++)  
        string[loop]=tolower(string[loop]);  
  
    printf("%s\n",string);  
    return 0;  
}
```